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Homework 6:

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S.4)

a) $A_c = \pi r_o^2$

$$r_o = \sqrt{A_c / \pi}$$

$$V = A_c \pi D$$

$$A_s = 2\pi r_o \pi D = 2\pi \sqrt{A_c / \pi} \pi D$$

character length: $L_c = V/A_s$

$$L_c = \frac{A_c \pi D}{2\pi \sqrt{A_c / \pi} \pi D}$$

$$= \frac{1}{2} \sqrt{\frac{A_c}{\pi}} = \frac{1}{2} \sqrt{\frac{7}{\pi}}$$

$$L_c = 0.74 \text{ mm} \quad 0.63 \text{ mm}$$

calculate the max center

$$L_c = r_o$$

$$= \sqrt{A_c / \pi} = \sqrt{\frac{7}{\pi}}$$

$$L_c = 1.49 \text{ mm} \quad 1.26 \text{ mm}$$

$$\text{Biot \#} = \frac{h L_c}{k} = \frac{50 (0.74 \times 10^{-3})}{2.3}$$

$$Bi_1 = 0.0162$$

$$Bi_2 = \frac{h L_c}{k} = \frac{50 (1.49 \times 10^{-3})}{2.3}$$

$$Bi_2 = 0.0323$$

Since $Bi_1, Bi_2 < 0.1$, lumped capacitance approx. is valid

b) $V = W L$

$$A_s = 2(W + w)L + 2Ww$$

$$L_c = \frac{WwL}{2(W + w)L + 2Ww}$$

$$L_c = \frac{7(5)(150)}{2(7+15)150 + 2(5)(7)}$$

$$L_c = 1.43 \text{ mm} \times 0.9 \text{ mm}$$

$$Bi = \frac{10 (1.43 \times 10^{-3})}{14.9} = 0.000959$$

$Bi < 0.1$, lumped capacitance approx is valid.

$$f) V = \frac{\pi D^2 L}{4}$$

$$A_s = \pi D L$$

$$L_c = \frac{\frac{\pi D^2 L}{4}}{\pi D L} = \frac{D}{4} = \frac{20}{4} = 5 \text{ mm}$$

$$L_c' = \frac{D}{2} = \frac{20}{2} = 10 \text{ mm}$$

$$h = \frac{PVC}{tA_s} \ln \frac{\theta_i}{\theta_o}$$

$$h = \frac{PVC}{4L} \ln \left(\frac{T_i - T_\infty}{T_o - T_\infty} \right)$$

$$h = \frac{(2300)(0.020)(1750)}{4(225)} \ln \left(\frac{200 - 20}{100 - 20} \right) = 72.5 \text{ W/m}^2 \cdot \text{K}$$

$$Bi = \frac{(72.5)(0.005)}{16} = 0.023, Bi = \frac{(72.5)(0.01)}{16} = 0.045$$

$Bi < 0.1$, lumped capacitance approx. is valid

$$g) L_{c,1} = \frac{V}{A_s} = \frac{D}{4} = \frac{200}{4} = 50 \text{ mm}$$

$$L_{c,2} = \frac{D}{2} = \frac{200}{2} = 100 \text{ mm}$$

$$h = \frac{PVC}{tA_s} \ln \frac{\theta_i}{\theta_o}$$

$$h = \frac{PVC}{4L} \ln \left(\frac{T_i - T_\infty}{T_o - T_\infty} \right)$$

$$h = \frac{(2300)(0.20)(1750)}{4(225)} \ln \left(\frac{200 - 20}{100 - 20} \right) = 725 \text{ W/m}^2 \cdot \text{K}$$

$$Bi = \frac{(725)(0.01)}{16} = 2.3, Bi = \frac{(725)(0.1)}{16} = 4.5$$

$Bi > 0.1$, lumped capacitance approx. is not valid

and does not have good

$$(6.3) \quad \frac{T - T_s}{T_\infty - T_s} = 1 - \exp\left(-Pr \frac{u_\infty y}{\nu}\right)$$

$$Pr = \frac{c_p \mu}{k} = 0.7$$

$$T_\infty = 400K, T_s = 300K, \frac{u_\infty}{\nu} = 5000m^{-1}$$

$$k = 0.0263 W/m \cdot K$$

(2.5)
(2.5)

$$\frac{\partial T}{\partial y} = (T_\infty - T_s) \exp\left[-Pr \frac{u_\infty y}{\nu}\right] \left(Pr \frac{u_\infty}{\nu}\right)$$

$$\frac{\partial T}{\partial y} \Big|_{y=0} = 100K (0.7 (5000m^{-1})) = 350,000 K/m$$

$$q'' = -k \frac{\partial T}{\partial y} \Big|_{y=0} = -0.0263 W/m \cdot K \cdot 350,000 K/m = \boxed{-9205 W/m^2}$$